

Md Salek Miah

Statistician — Epidemiology & Public Health — Machine Learning & Data Science

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🌐 Portfolio

🌐 LinkedIn — 🐙 GitHub — 📄 ResearchGate — 🏠 Google Scholar — 📄 ORCID

RESEARCH PROFILE

Skilled in statistical modeling and survey data analysis, eager to collaborate and learn new research methodologies in machine learning, data science, and spatial epidemiology.

SELECTED PUBLICATIONS

Published Q1 Journal Articles:

- **Associations of Internet Use and Pregnancy Loss with Depression and Anxiety among Women in Bangladesh: Evidence from the 2022 BDHS**
BMC Women's Health
(Published 29 Nov 2025, DOI: [10.1186/s12905-025-04166-4](https://doi.org/10.1186/s12905-025-04166-4))

Under Review (Q1 Journals, Scientific Reports, BMC etc.):

- **Machine Learning Analysis of Factors Influencing Skilled Birth Attendance in Burkina Faso: Assessing Spatial Inequalities Using Imbalanced Survey Data**
Scientific Reports (Q1)
(Under Review)
- **Exploring Determinants and Spatial Inequalities of Quality Antenatal Care in Nepal Using Machine Learning on Complex Survey Data**
Scientific Reports (Q1)
(Under Review)
- **Identifying Key Determinants and Spatial Insights of Skilled Birth Assistance in Bangladesh: A Machine Learning Approach from BDHS 2022**
Scientific Reports (Q1)
(Under Review)
- **A Comparative Study of Machine Learning Algorithms in Predicting and Identifying Risk Factors of Skilled Birth Assistance in Nepal**
BMC Women's Health (Q1)
(Under Review)
- **Associations of Women's Empowerment and Intimate Partner Violence with Mental Health Among Ever-Married Women in Bangladesh**
Archives of Women's Mental Health (Q1)
(Under Review)

Preprints:

- **Determinants and Spatial Variation of Skilled Birth Assistance in Nepal: Insights from NDHS 2022**
Research Square
DOI: [10.21203/rs.3.rs-7973803/v1](https://doi.org/10.21203/rs.3.rs-7973803/v1) — CC BY 4.0 License
- **Associations of Women's Empowerment and Intimate Partner Violence with Mental Health Among Ever-Married Women in Bangladesh**
Research Square
DOI: [10.21203/rs.3.rs-7473860/v1](https://doi.org/10.21203/rs.3.rs-7473860/v1)

RESEARCH EXPERIENCE

Research Assistant — Biostatistics, Epidemiology & Public Health Research Team, SUST, Sylhet, Bangladesh

- Worked under supervision of Dr. Md Jamal Uddin (Professor of Biostatistics and Epidemiology) and Dr. Moham-mad Ohid Ullah (Professor, Department of Statistics, SUST) as senior researchers.
- Contributed to 10 manuscripts currently under review in **WoS Q1 journals** in public health, epidemiology, and machine learning applications.
- Assisted in statistical analyses, machine learning modeling, and data interpretation of national survey datasets.
- Conducted spatial analysis, data management, and manuscript preparation.

Current Research Project — Impact of Pesticides Use on Farmers’ Health, North-Eastern Bangladesh (2025–26)

- Principal Investigator: Dr. Mohammad Ohid Ullah, Professor, Department of Statistics, SUST
- Role: Research Assistant supporting senior researchers in data collection, statistical analysis, and report preparation
- Funded by: SUST Research Centre

STATISTICS & MACHINE LEARNING SKILLS

Multilevel Modeling, Survival Analysis, Structural Equation Modeling, Complex Survey Analysis, Epidemiology
Machine Learning Algorithm

- **Data Tools:** R Studio , Python, STATA, SPSS, MySQL, ArcGIS

PROJECTS

- **Interactive Statistical Calculator** – Developed using R Shiny. [Live Demo](#)
- **MovieLens Recommendation System** – Developed an interactive movie recommendation application using R Studio and Hugging Face Spaces. [Live Demo](#)

EDUCATION

Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh
B.Sc. (Hons.) in Statistics 2021–Present

AWARDS

- Winner, DNA Day Writing Contest – ISCB RSG Bangladesh, 2025
- Top 10 Performer, Data Hackathon – Comilla University, 2025

CERTIFICATIONS

Machine Learning Certifications:

- **Data Science and Machine Learning with Python**
Instructor: Coursera — Issued Jan 2025
Covers data preprocessing, analysis, and visualization using Pandas, NumPy, Matplotlib. [Verified Certificate Link](#)
- **Data Analysis with R**
IBM via Coursera — Issued Oct 2024 — Validation ID: PWID-B0442600
Covers data wrangling, EDA, and model evaluation. [Verified Certificate Link](#)
- **Exploratory Data Analysis for Machine Learning**
IBM — Issued Jan 2025 — 14 hours
Covers data exploration techniques for ML. [Verified Certificate Link](#)

PROFESSIONAL AFFILIATIONS

- Research Collaborator, Biostatistics & Epidemiology Research Team, SUST
- Member, International Society for Computational Biology (ISCB)
- Member, Bangladesh Statistical Association (BSA)
- Member, Daffodil International University Research Society

References available upon request (including Dr. Mohammad Ohid Ullah, Dr. Md Jamal Uddin)